R101/B | Two-stage provenance

Fixed TRAIN hint endpoint, then independently reset original full18 learning.

Teacher: pretrained. Stage I:12800 updates, logical batch1024, seed2; CVRP+15MVRP TRAIN81000/task only.

Separate non-affine per-node LayerNorm MSE; one dropout forward. No outcome labels, learned adapter or teacher VAL/TEST. Fixed endpoint, not validation selected.

Stage II: fresh original R84/R88B student; ONLY instance_encoder transferred. All non-encoder parameters remain identical fresh R84. Optimizer, scaler, RNG and task/sample streams reset.

Original1842953 parameters; no added heads or features. R77 CE .35 / comparison .35 / risk .02 / R-Drop .35. No hint loss, teacher arrays or online donor in Stage II or deployment.

Both A/B have12800+57600 updates; R89B/R84A are NOT total-budget-matched. B teacher has additional external solver pretraining history. The paper memo provides conditional motivation only.

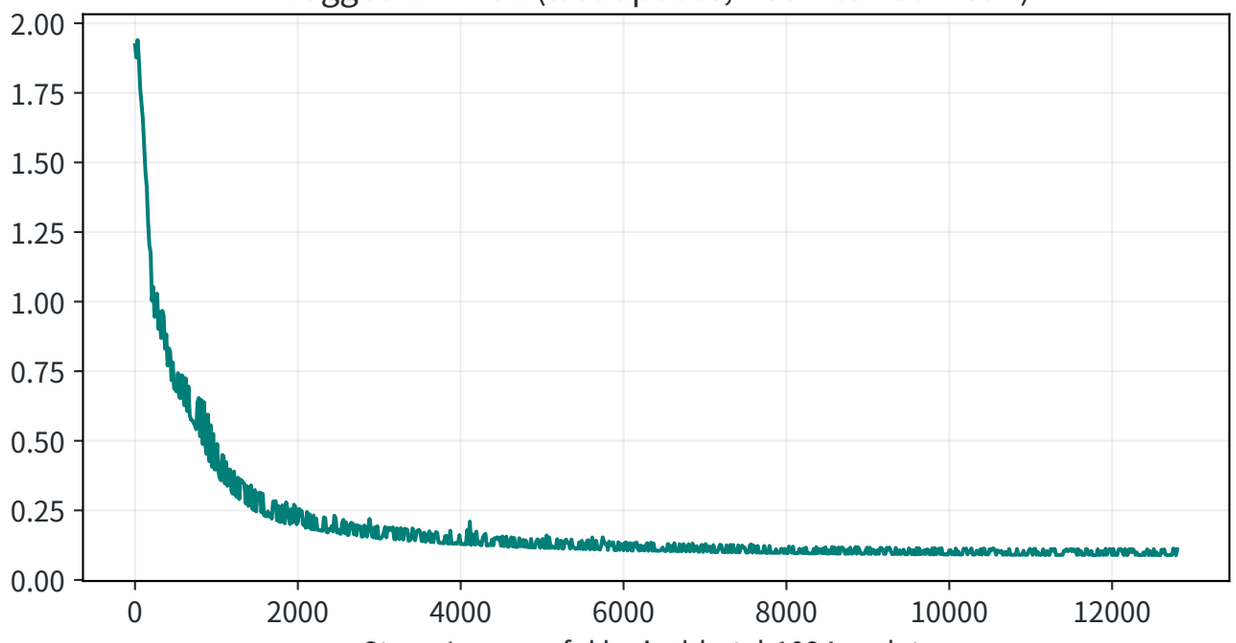
Native/FP64 ind winners; legal-pool Oracle and TRAIN-fitted SBS; ALL is task-equal. Development-exposed TEST; no TEST/LIB checkpoint selection.

Stage-I encoder SHA256: 6461d4d78289f6cbb92f534034602eea4eba389fc606f1fc0b386ed2662c657a

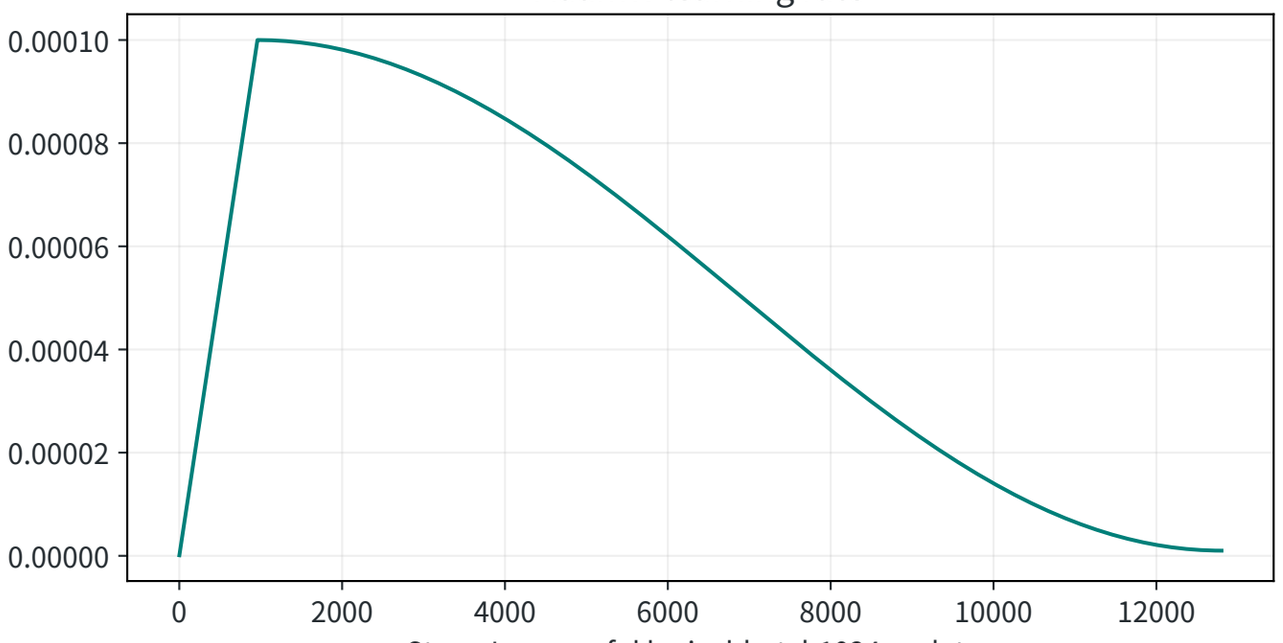
Unchanged non-encoder SHA256: 970a8aa2a548f9034f34b73d060662f71808e46359335185ce07bfc5b1bb2fa6

R101/B | TRAIN-only pretrained node hints; fixed12800 endpoint

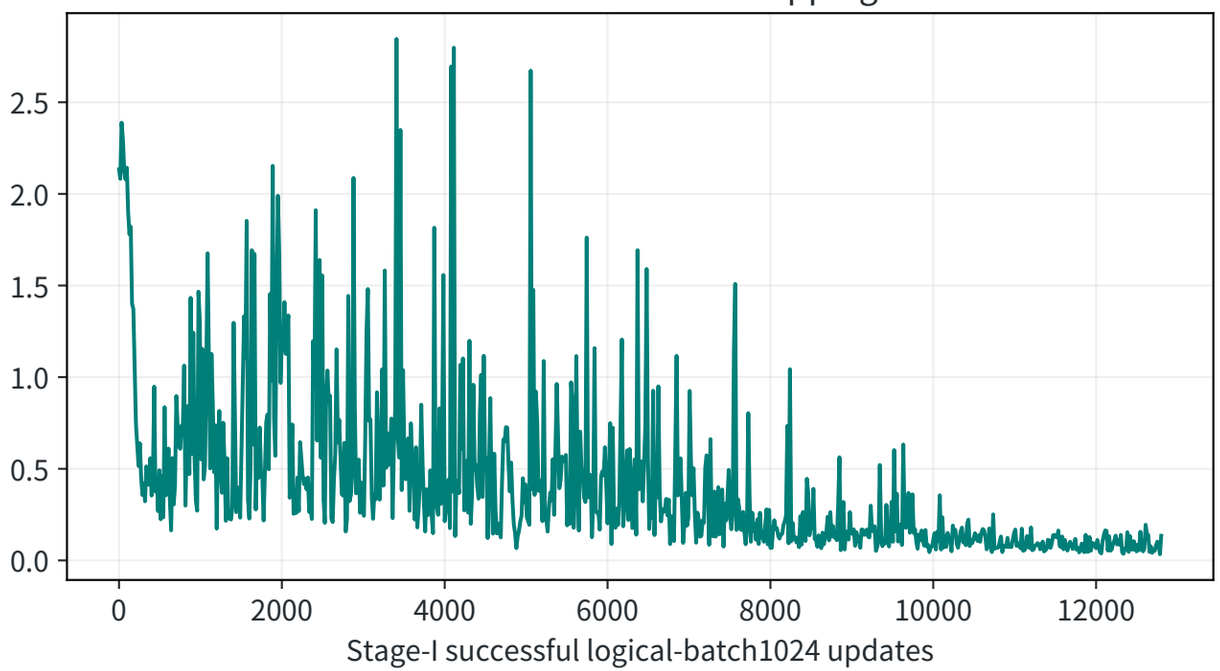
Logged LN-MSE (last update, not interval mean)



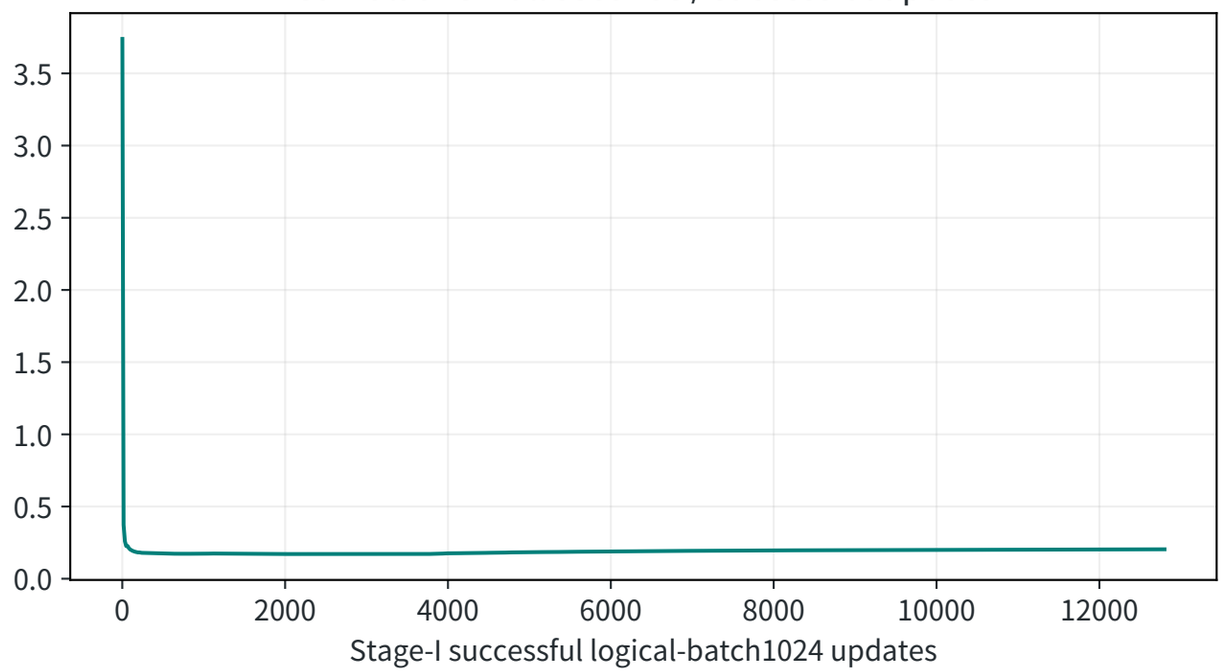
AdamW learning rate



Gradient norm before clipping



Cumulative wall seconds / successful update



R101/B | Locked full18 results

Same unique VAL checkpoint on VAL/TEST/LIB; native/FP64 ind; historical NSS weights disclosed.

epoch 27 | update 38880 | seed2 | fresh initialization

Data: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
VAL ALL	71.2012	66.2037	11.416300	11.427978	11.371692	-0.1965	+0.3911	0.4000
TEST ALL	71.4346	66.1802	11.373450	11.385299	11.329932	-0.2000	+0.3831	0.3910

Dataset	n	NSS Top1 (%)	R101 Top1 (%)	NSS cost	R101 cost
TSP TEST	4500	45.888889	47.400000	7.970400719	7.970903724
CVRP TEST	4500	45.977778	47.822222	18.004802368	18.008762029
TSPLIB	49	20.408163	30.612245	8.217907074	8.195489915
CVRPLIB	100	70.000000	63.000000	67.101111717	67.248113565

NSS: R79 weights trained on historical70114 TRAIN, selected using historical VAL; not retrained on new90k.

NSS TEST predictions were freshly computed on these4500 instances; old3895 TEST metrics are not reused.

SBS is fitted on the new81000 TRAIN only. Oracle is the per-instance legal-pool minimum.

ALL is the equal-problem mean; mixed raw cost is descriptive. TEST informed prior research.

Method selection and true winner use separate colors on the same axes; Top1 is per-instance accuracy.

Selection head: original_score_mlp_maximin | objective: original R77 native CE

R101/B | VAL: ALL and18 tasks

TRAIN-fitted SBS and unchanged native/FP64 costs; no per-task checkpoint assembly.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)	TRAIN SBS
ALL	71.2012	66.2037	11.416300	11.427978	11.371692	-0.1965	+0.3911	0.4000	per-problem TRAIN SBS
TSP	47.0667	33.1111	8.024764	8.045354	7.990533	-0.2559	+0.4284	0.4168	T2T
CVRP	46.8444	18.7111	18.157905	18.269377	18.064087	-0.6102	+0.5194	0.6885	RELD_CVRP
ATSP	76.9778	47.4889	1.638750	1.676905	1.632471	-2.2753	+0.3847	0.3807	UNICO_MatPOENet
OVRP	41.1556	35.7556	7.461777	7.469579	7.415777	-0.1044	+0.6203	0.6360	MoSES_RF
VRPB	81.1778	80.8444	9.354736	9.354876	9.318750	-0.0015	+0.3862	0.3861	reld_mt
VRPL	38.0444	33.0889	12.431904	12.437959	12.383444	-0.0487	+0.3913	0.3992	MoSES_RF
VRPTW	80.1333	79.9556	18.818619	18.819431	18.766139	-0.0043	+0.2797	0.2950	reld_mt
OVRPTW	83.8889	83.7778	10.143708	10.143730	10.116859	-0.0002	+0.2654	0.2643	reld_mt
OVRPB	88.5556	88.5111	6.357796	6.357254	6.341078	+0.0085	+0.2636	0.2777	reld_mt
OVRPL	41.6667	35.9111	7.437495	7.445709	7.390416	-0.1103	+0.6370	0.6536	MoSES_RF
VRPBL	78.9333	78.5111	9.367756	9.371523	9.328366	-0.0402	+0.4223	0.4238	reld_mt
VRPBTW	78.7778	78.4222	19.945192	19.950430	19.851084	-0.0263	+0.4741	0.4523	reld_mt
VRPLTW	78.9333	78.7556	18.763489	18.765871	18.707843	-0.0127	+0.2974	0.3115	reld_mt
OVRPBL	89.4222	89.2000	6.368941	6.370093	6.352824	-0.0181	+0.2537	0.2633	reld_mt
OVRPBTW	83.6667	83.4444	10.659787	10.661098	10.623245	-0.0123	+0.3440	0.3123	reld_mt
OVRPLTW	83.6222	83.4889	10.156412	10.157423	10.129818	-0.0100	+0.2625	0.2638	reld_mt
VRPBLTW	79.4444	79.4000	19.794736	19.797044	19.706420	-0.0117	+0.4482	0.4360	reld_mt
OVRPBLTW	83.3111	83.2889	10.609625	10.609955	10.571302	-0.0031	+0.3625	0.3389	reld_mt

vs_SBS=100*(mean_cost/SBS_cost-1); vs_Oracle=100*(mean_cost/Oracle_cost-1).

actual regret=mean_i[100*(selected_i-oracle_i)/oracle_i]; not a ratio of means.

TRAIN81000 / VAL4500 / TEST4500 per problem. All18 problems have equal macro weight.

R101/B | TEST: ALL and18 tasks

TRAIN-fitted SBS and unchanged native/FP64 costs; no per-task checkpoint assembly.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)	TRAIN SBS
ALL	71.4346	66.1802	11.373450	11.385299	11.329932	-0.2000	+0.3831	0.3910	per-problem TRAIN SBS
TSP	47.4000	34.5333	7.970904	7.989621	7.937593	-0.2343	+0.4197	0.4137	T2T
CVRP	47.8222	17.9333	18.008762	18.124983	17.918509	-0.6412	+0.5037	0.6601	RELD_CVRP
ATSP	76.8889	47.2444	1.635283	1.674207	1.629441	-2.3249	+0.3585	0.3578	UNICO_MatPOENet
OVRP	41.4889	33.6444	7.445720	7.455750	7.400487	-0.1345	+0.6112	0.6191	MoSES_RF
VRPB	80.7333	80.4000	9.349268	9.351810	9.315298	-0.0272	+0.3647	0.3696	reld_mt
VRPL	37.9556	32.2667	12.349619	12.356138	12.300958	-0.0528	+0.3956	0.4030	MoSES_RF
VRPTW	81.1333	80.8444	18.688695	18.690545	18.637408	-0.0099	+0.2752	0.2839	reld_mt
OVRPTW	83.9778	83.9333	10.092510	10.093063	10.068135	-0.0055	+0.2421	0.2329	reld_mt
OVRPB	88.0667	88.1333	6.371061	6.370826	6.352225	+0.0037	+0.2965	0.3069	reld_mt
OVRPL	41.4667	34.6000	7.464103	7.472542	7.418033	-0.1129	+0.6211	0.6301	MoSES_RF
VRPBL	79.6667	79.4222	9.395508	9.396379	9.355967	-0.0093	+0.4226	0.4182	reld_mt
VRPBTW	79.0222	78.7556	19.662672	19.665725	19.577666	-0.0155	+0.4342	0.4307	reld_mt
VRPLTW	79.4222	79.2444	18.801794	18.803136	18.745143	-0.0071	+0.3022	0.3203	reld_mt
OVRPBL	89.2444	89.2889	6.375857	6.375741	6.359310	+0.0018	+0.2602	0.2736	reld_mt
OVRPBTW	84.9556	84.8222	10.658871	10.659591	10.624748	-0.0068	+0.3212	0.2930	reld_mt
OVRPLTW	82.9778	82.8667	10.085927	10.086112	10.059001	-0.0018	+0.2677	0.2681	reld_mt
VRPBLTW	79.8667	79.7333	19.704221	19.707068	19.613284	-0.0144	+0.4637	0.4440	reld_mt
OVRPBLTW	83.7333	83.5778	10.661334	10.662149	10.625574	-0.0076	+0.3366	0.3134	reld_mt

vs_SBS=100*(mean_cost/SBS_cost-1); vs_Oracle=100*(mean_cost/Oracle_cost-1).

actual regret=mean_i[100*(selected_i-oracle_i)/oracle_i]; not a ratio of means.

TRAIN81000 / VAL4500 / TEST4500 per problem. All18 problems have equal macro weight.

R101/B | LIB and locked provenance

Current R101 factory, same buffers and selected checkpoint; no solver execution or LIB tuning.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
TSPLIB	30.6122	10.2041	8.195490	8.240435	8.142167	-0.5454	+0.6549	0.6673
CVRPLIB	63.0000	45.0000	67.248114	67.245364	66.945093	+0.0041	+0.4526	0.5681

SHA256: cd1f5e55d852c2460d3949825f1a7f923b685295146e93a542c753611ef0683b

Unique selection: strict minimum full VAL macro_vs_sbs_pct with common new81000 TRAIN-fitted SBS; first wins ties

Inference: eval mode, original R88B physical TSP node0 slot3; original R77 hand34/methodID, joint2, score MLP/pair differences/maximin; stable ascending global-solver-ID ties; original inputs only; no teacher, hint, adapter or calibration

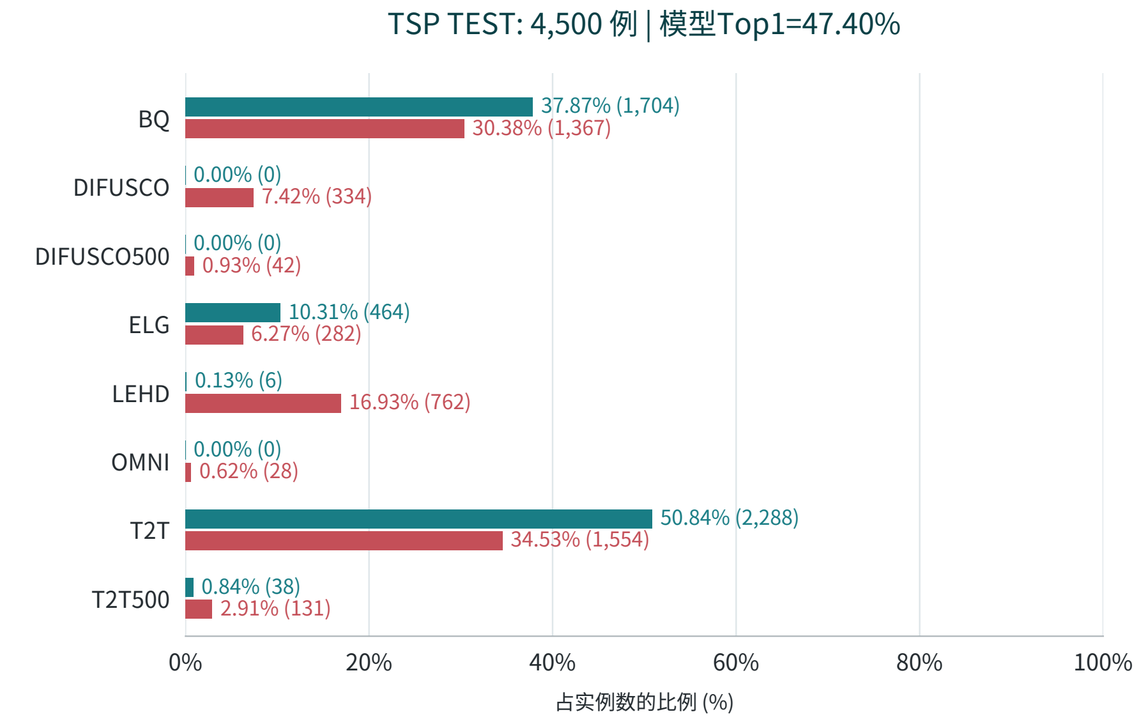
Normalization: new81000 TRAIN only; no TEST/LIB refit.

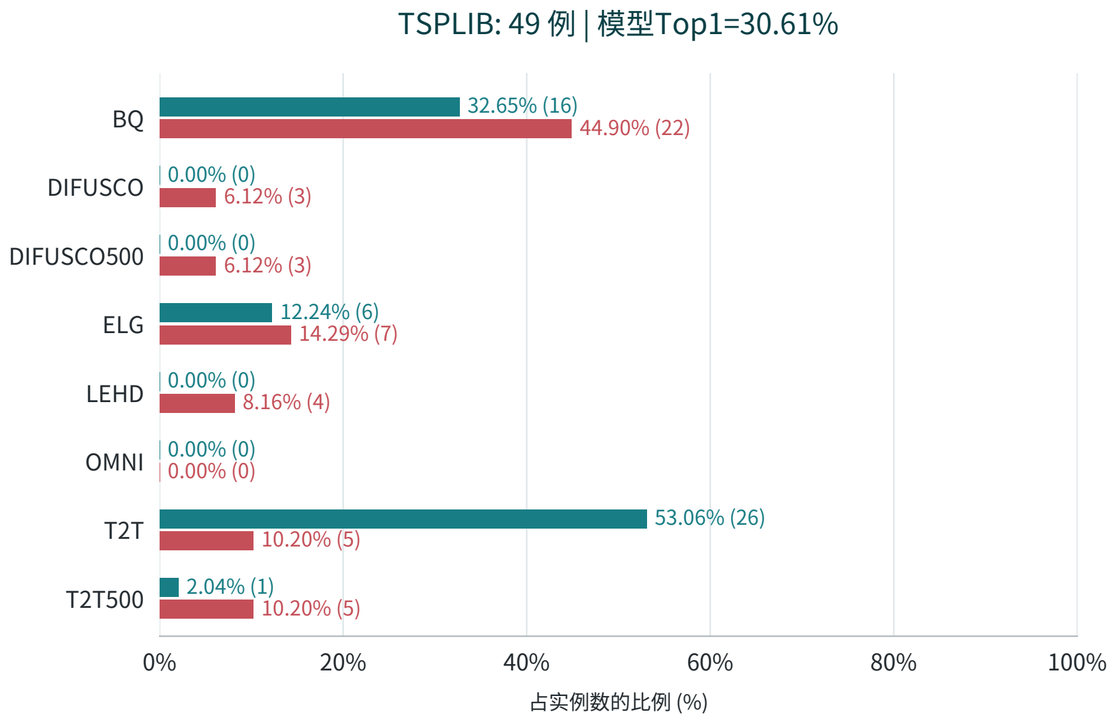
R101/B | TSP | 方法分布

同一锁定模型: epoch 27 / update 38880 / SHA256 cd1f5e55d852c246

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

LIB: data/TSPLIB | native/FP64 ind





标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

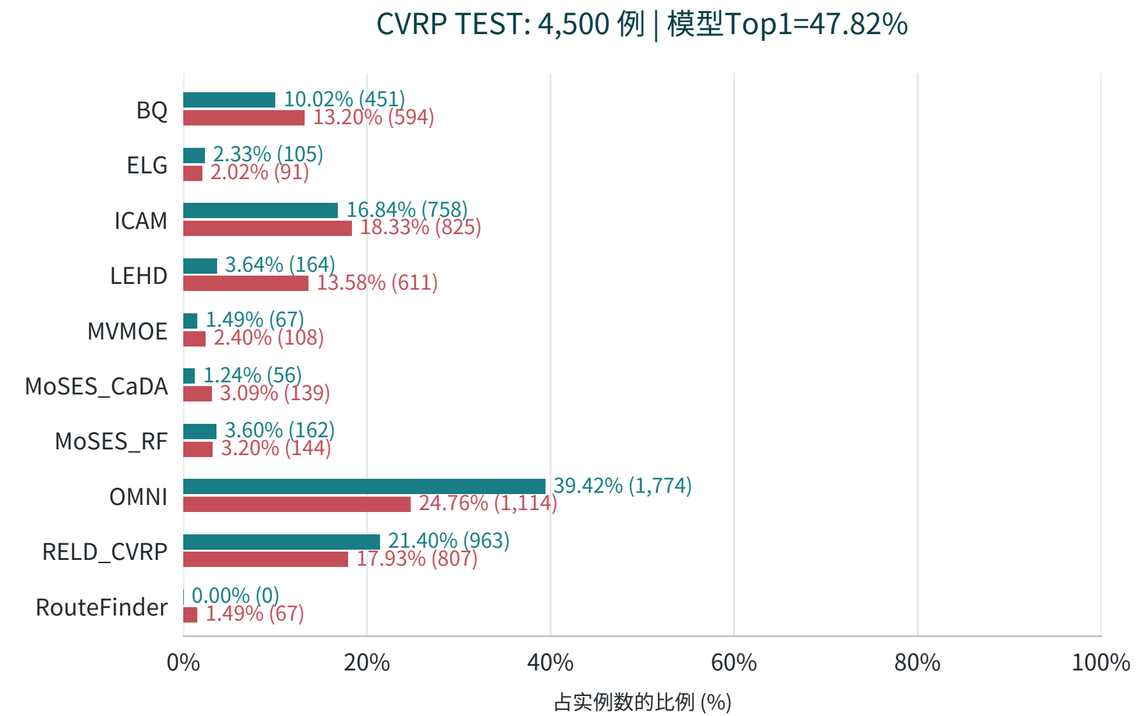
R101/B: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

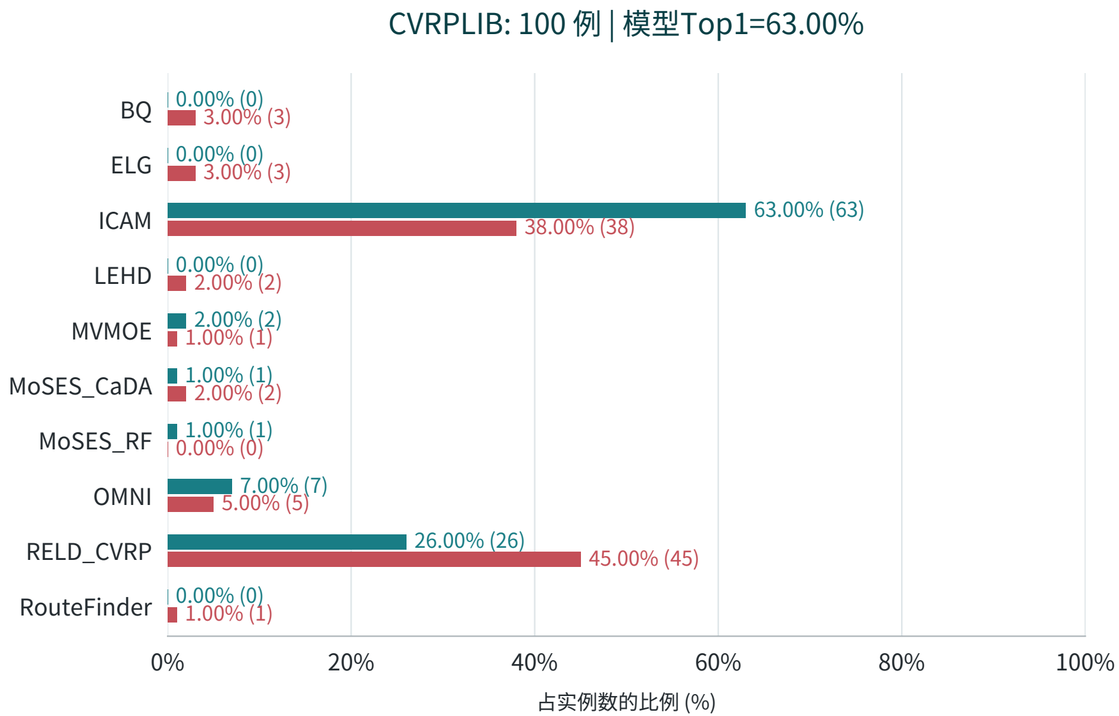
R101/B | CVRP | 方法分布

同一锁定模型: epoch 27 / update 38880 / SHA256 cd1f5e55d852c246

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

LIB: data/CVRPLIB | native/FP64 ind





标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

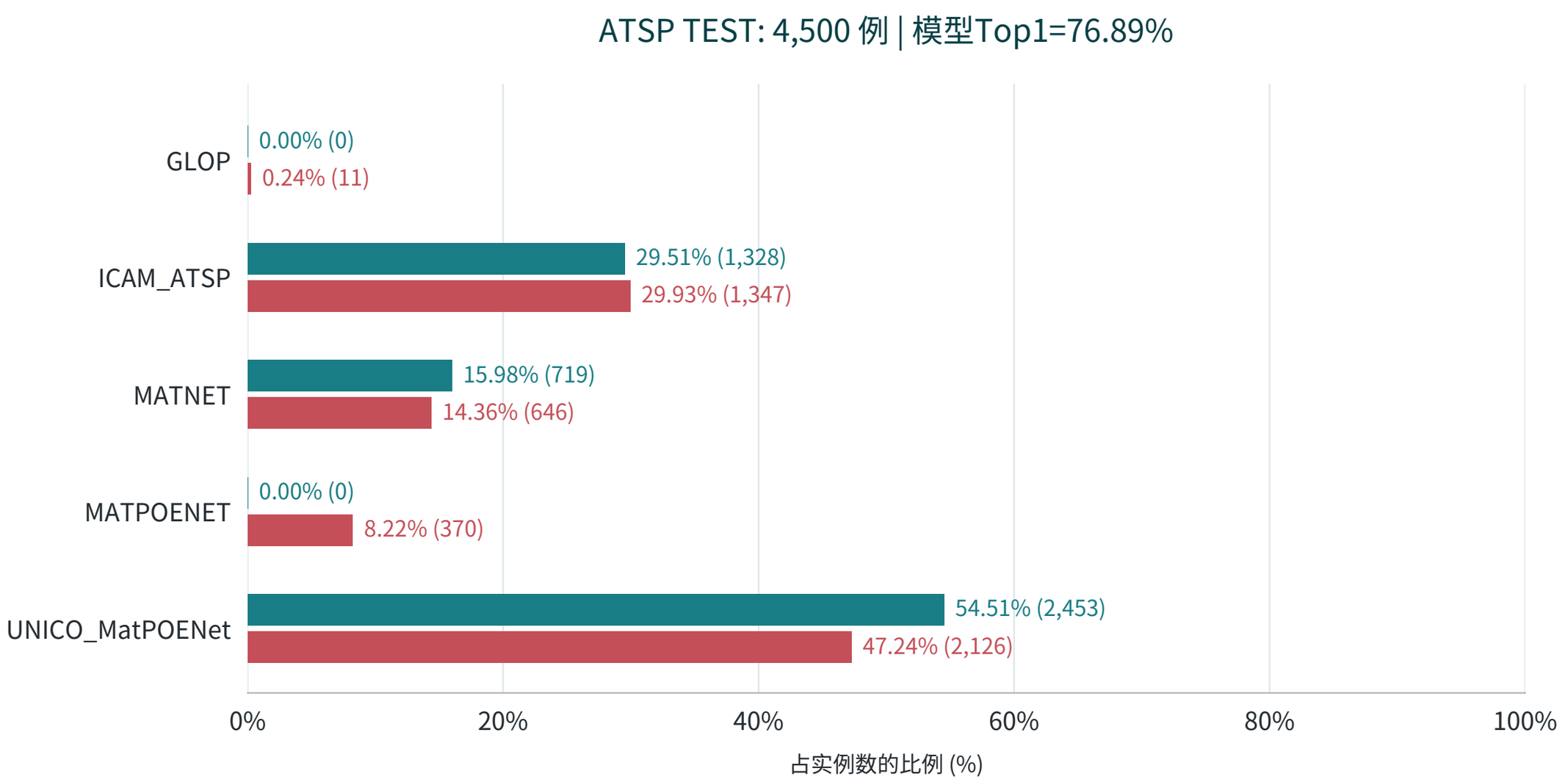
R101/B: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/B | ATSP | 方法分布

同一锁定模型: epoch 27 / update 38880 / SHA256 cd1f5e55d852c246

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

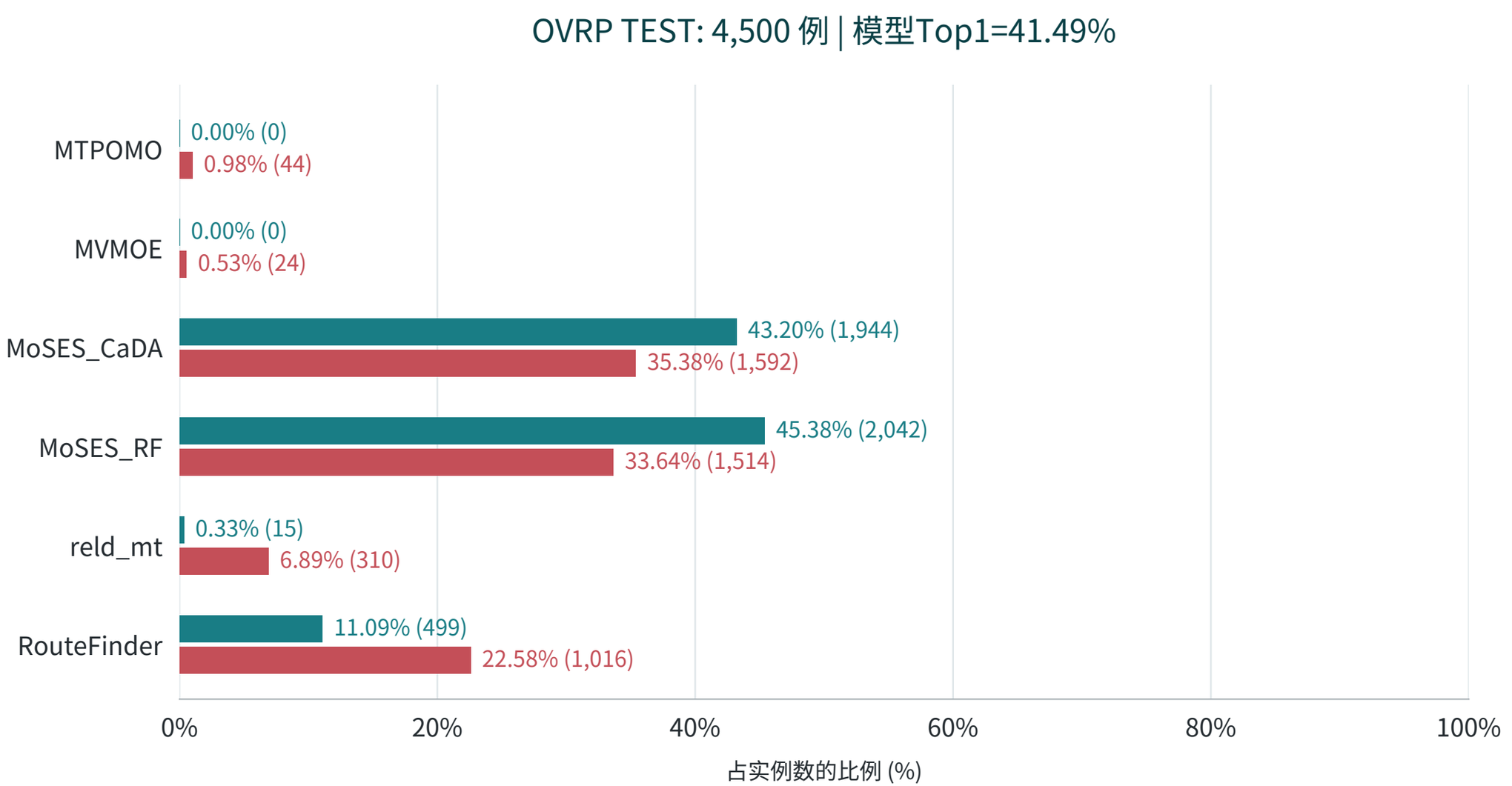
R101/B: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/B | OVRP | 方法分布

同一锁定模型: epoch 27 / update 38880 / SHA256 cd1f5e55d852c246

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

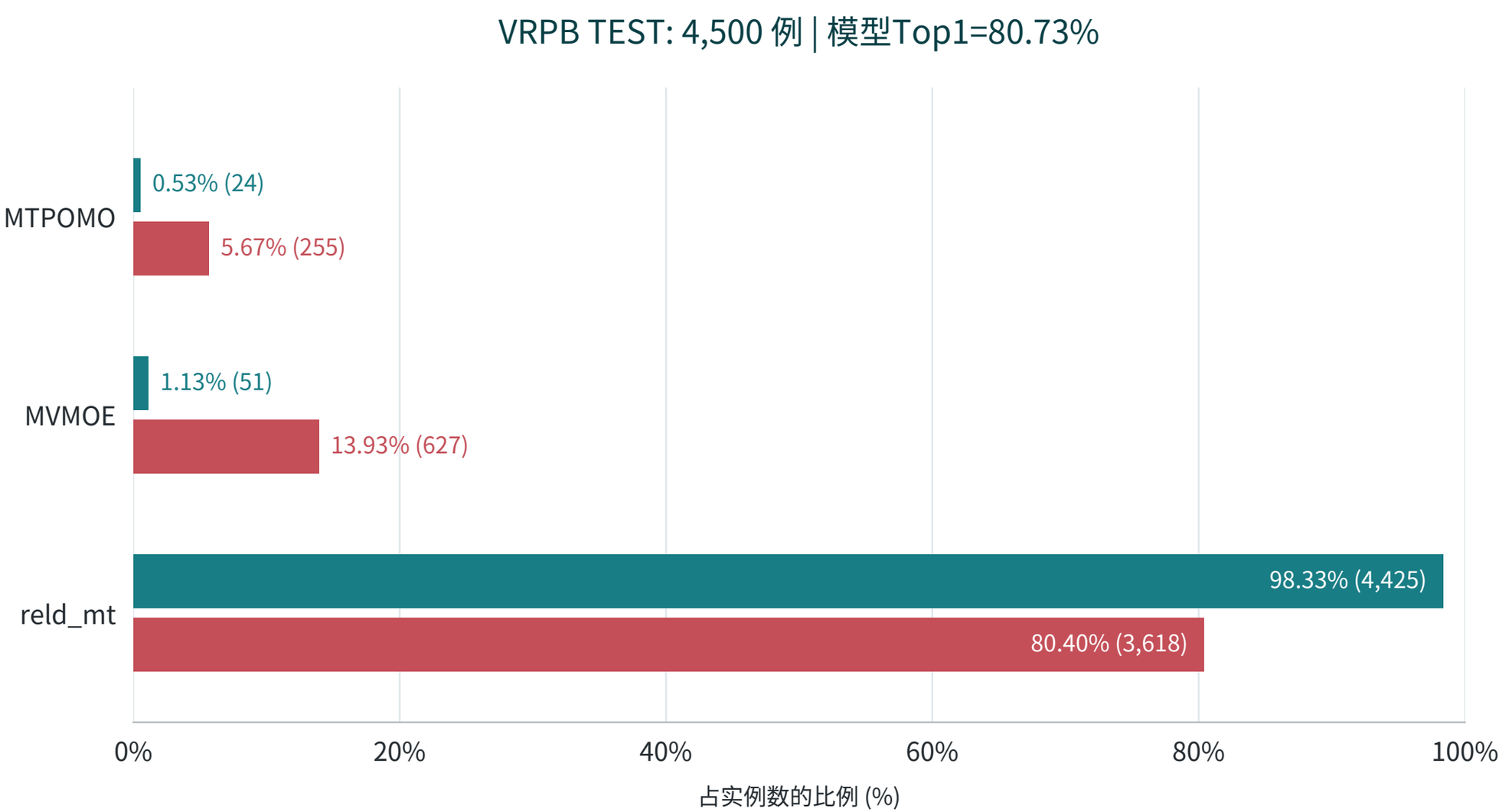
R101/B: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/B | VRPB | 方法分布

同一锁定模型: epoch 27 / update 38880 / SHA256 cd1f5e55d852c246

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

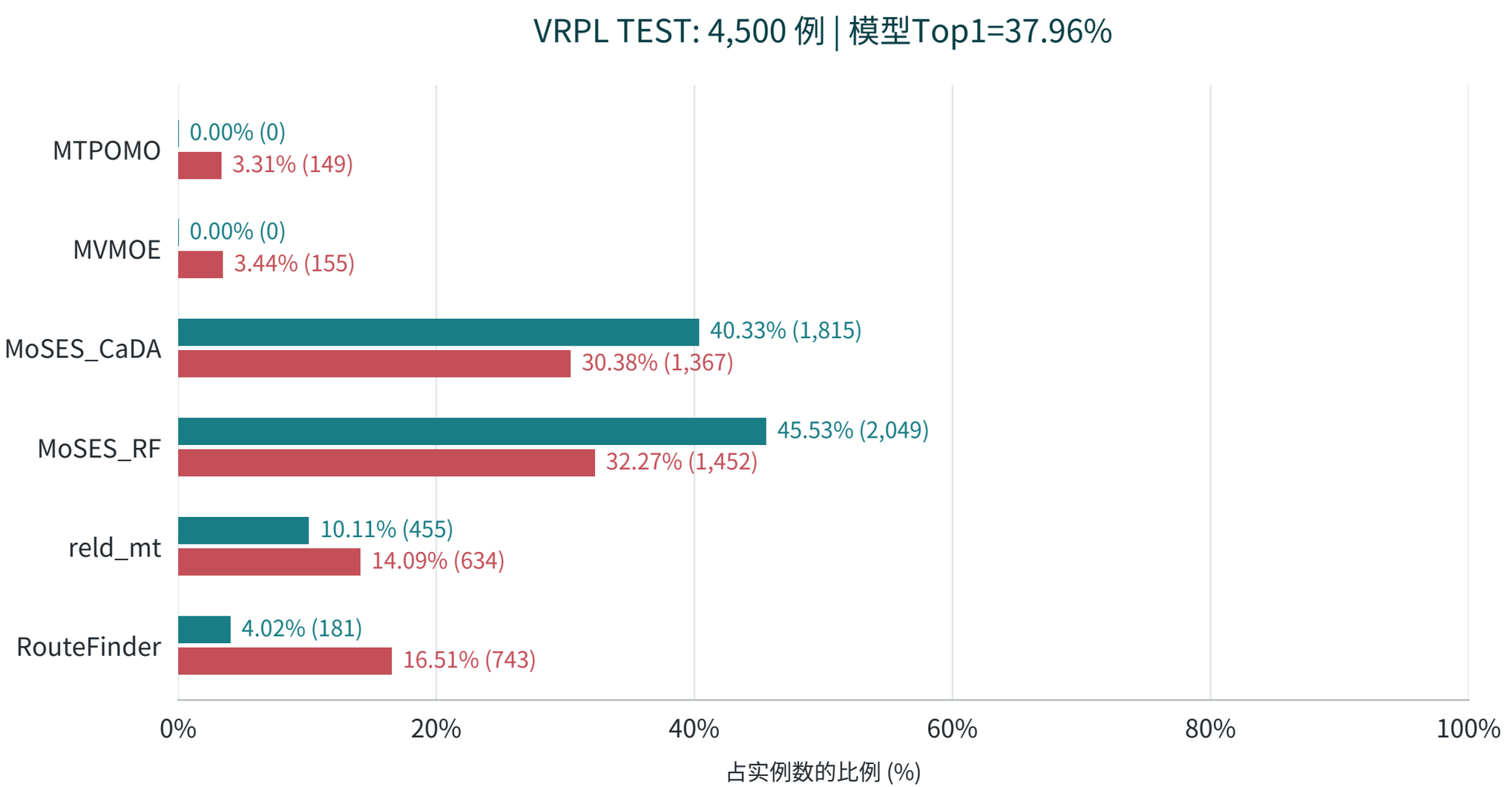
R101/B: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/B | VRPL | 方法分布

同一锁定模型: epoch 27 / update 38880 / SHA256 cd1f5e55d852c246

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

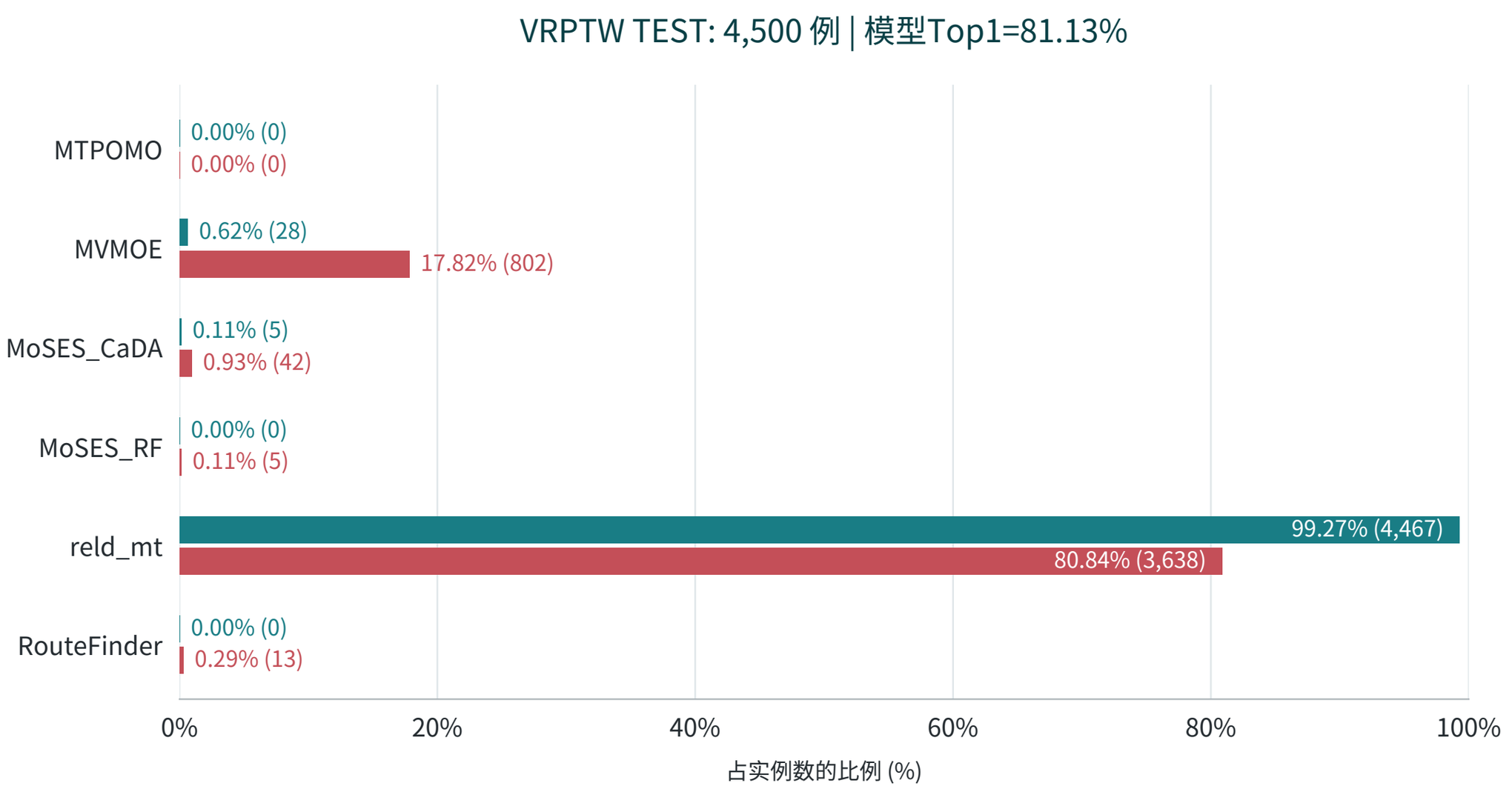
R101/B: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/B | VRPTW | 方法分布

同一锁定模型: epoch 27 / update 38880 / SHA256 cd1f5e55d852c246

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

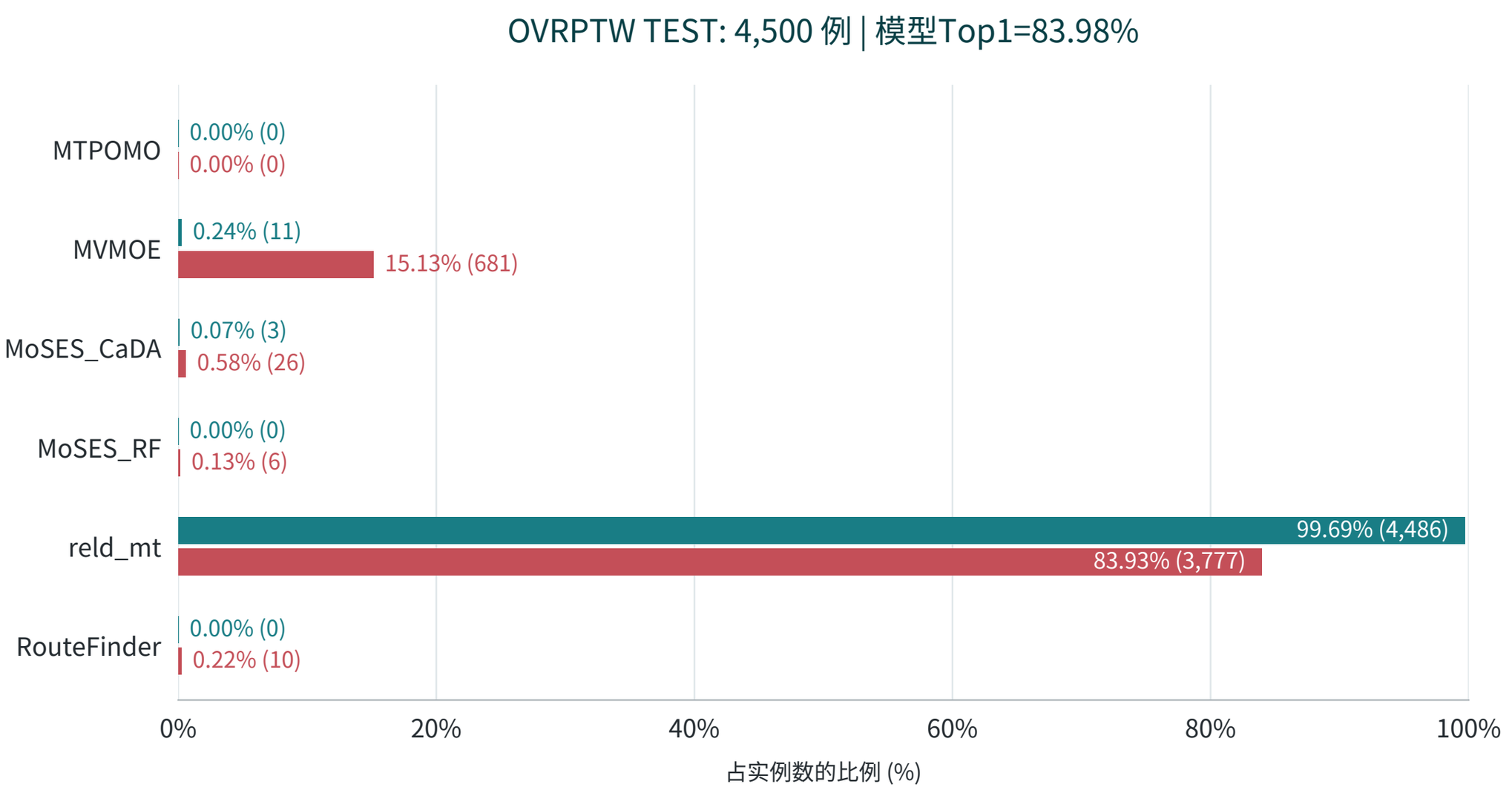
R101/B: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/B | OVRPTW | 方法分布

同一锁定模型: epoch 27 / update 38880 / SHA256 cd1f5e55d852c246

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

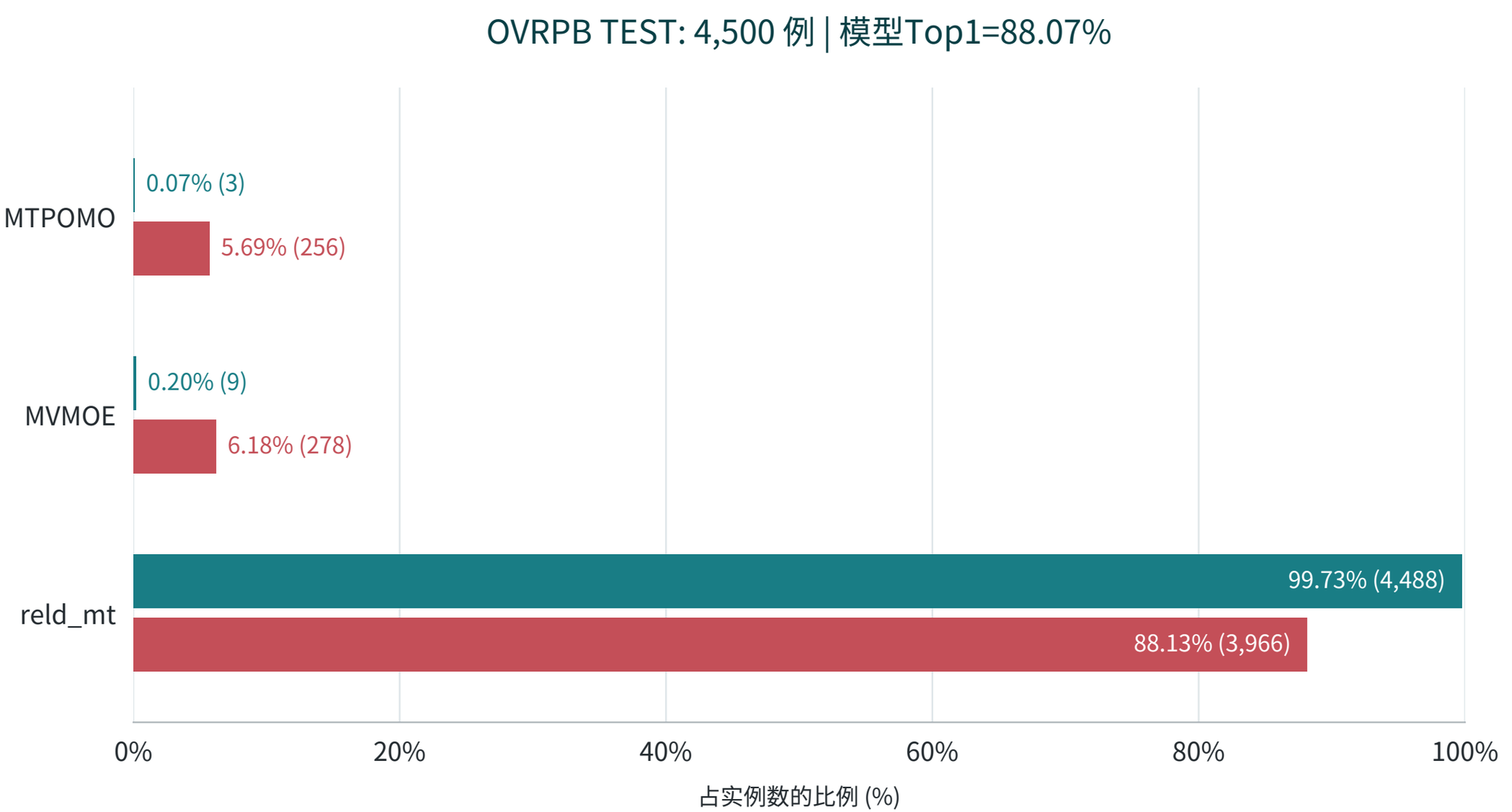
R101/B: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/B | OVRPB | 方法分布

同一锁定模型: epoch 27 / update 38880 / SHA256 cd1f5e55d852c246

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

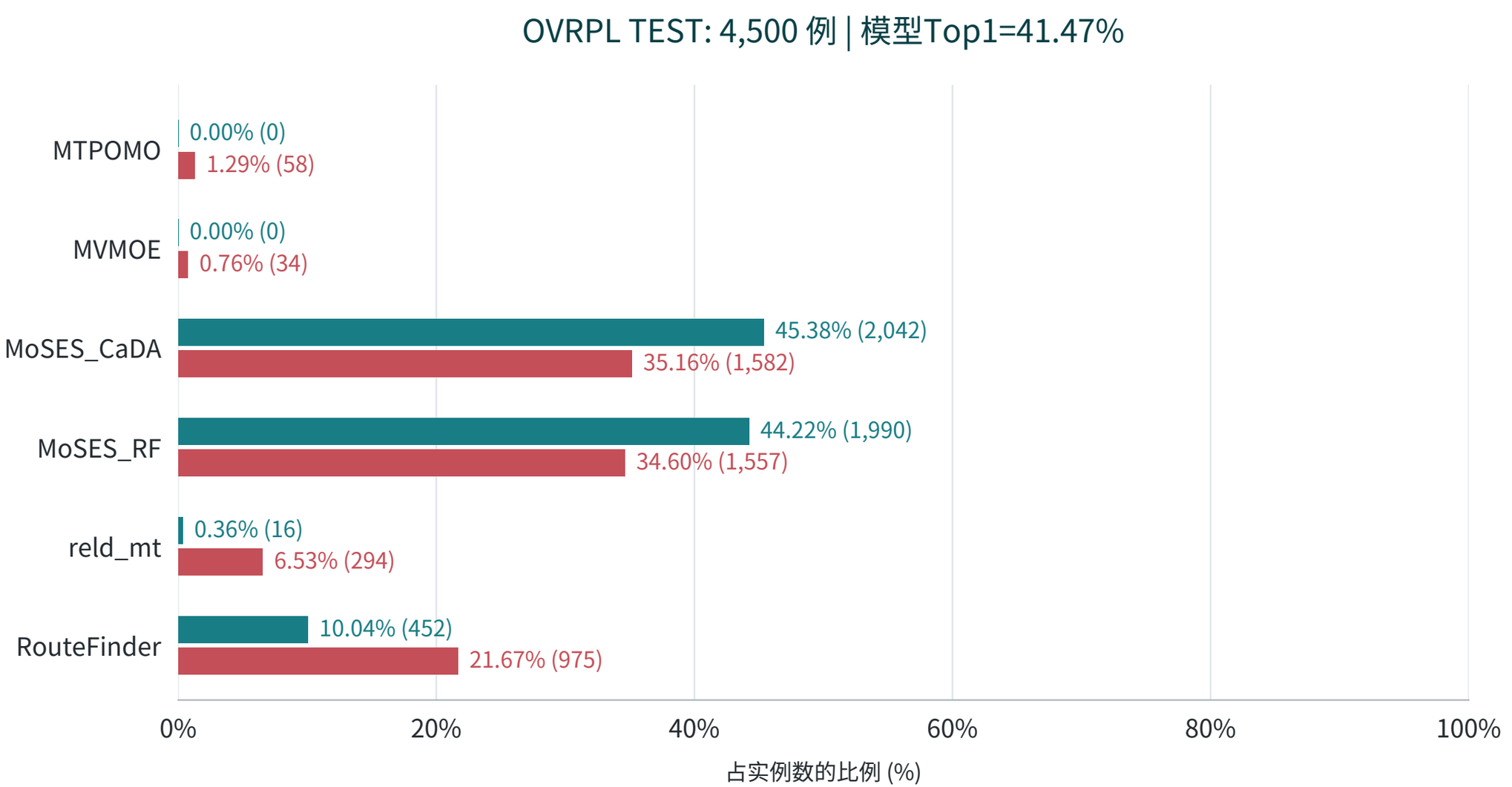
R101/B: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/B | OVRPL | 方法分布

同一锁定模型: epoch 27 / update 38880 / SHA256 cd1f5e55d852c246

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

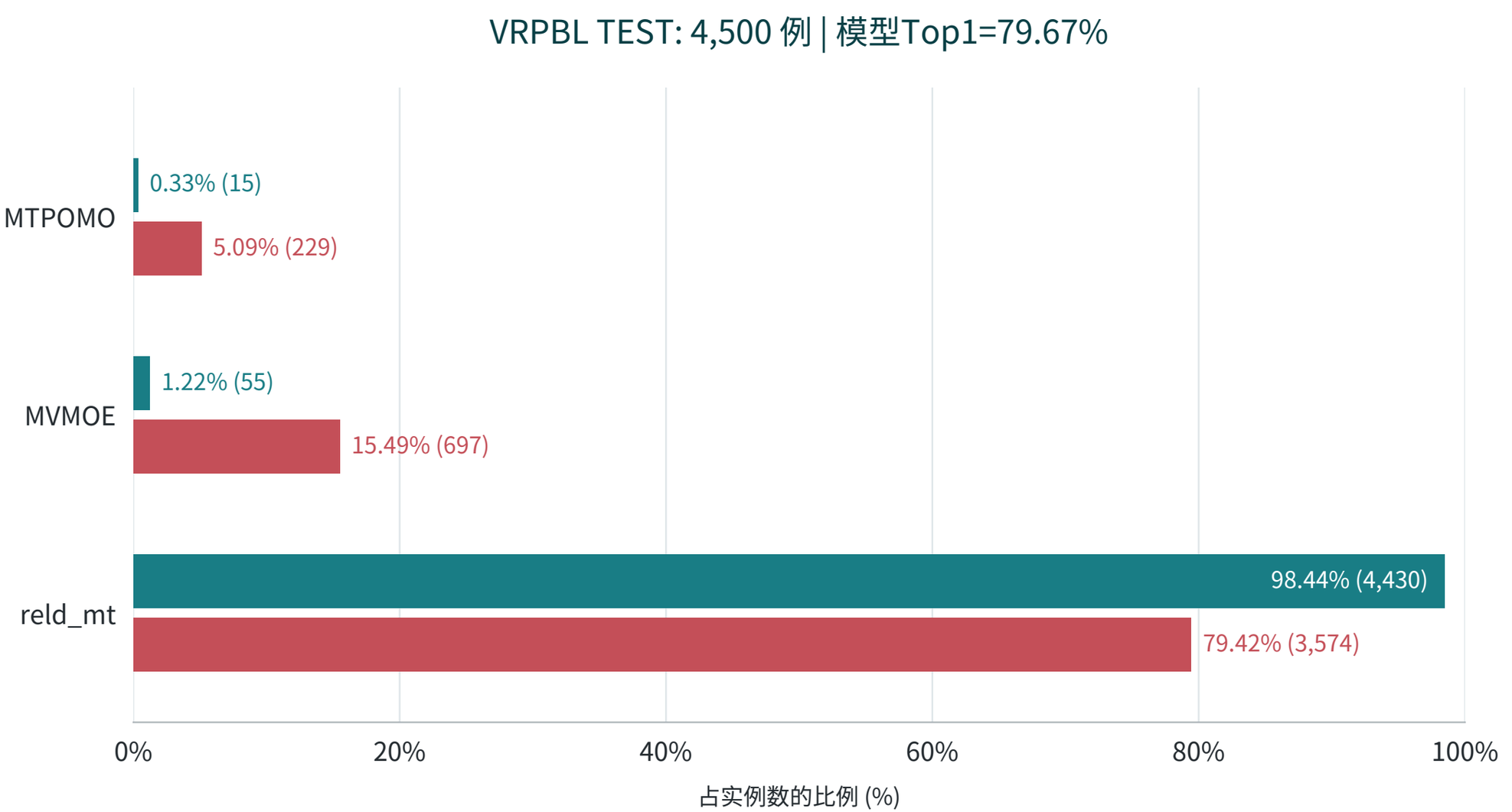
R101/B: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/B | VRPBL | 方法分布

同一锁定模型: epoch 27 / update 38880 / SHA256 cd1f5e55d852c246

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

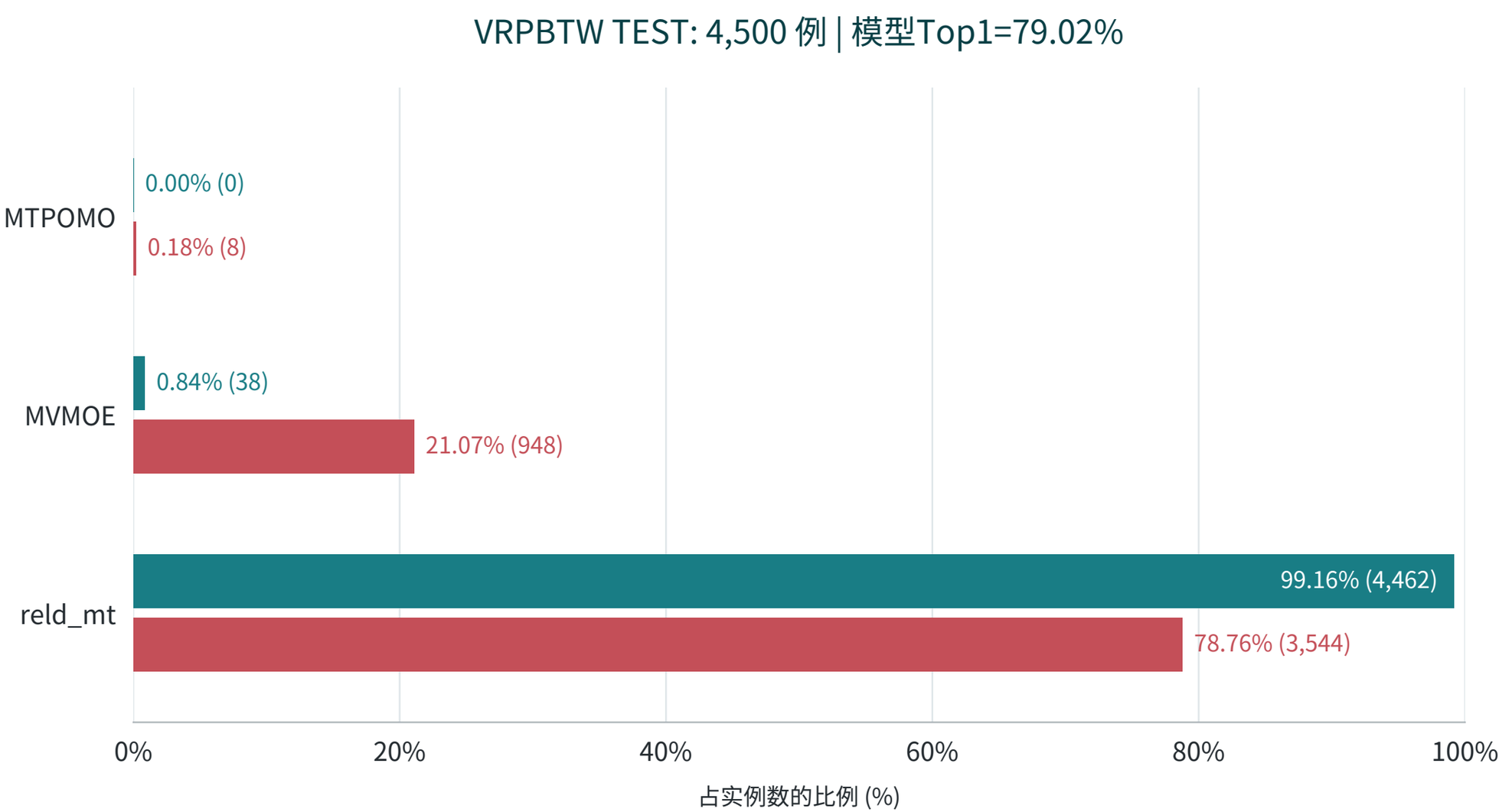
R101/B: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/B | VRPBTW | 方法分布

同一锁定模型: epoch 27 / update 38880 / SHA256 cd1f5e55d852c246

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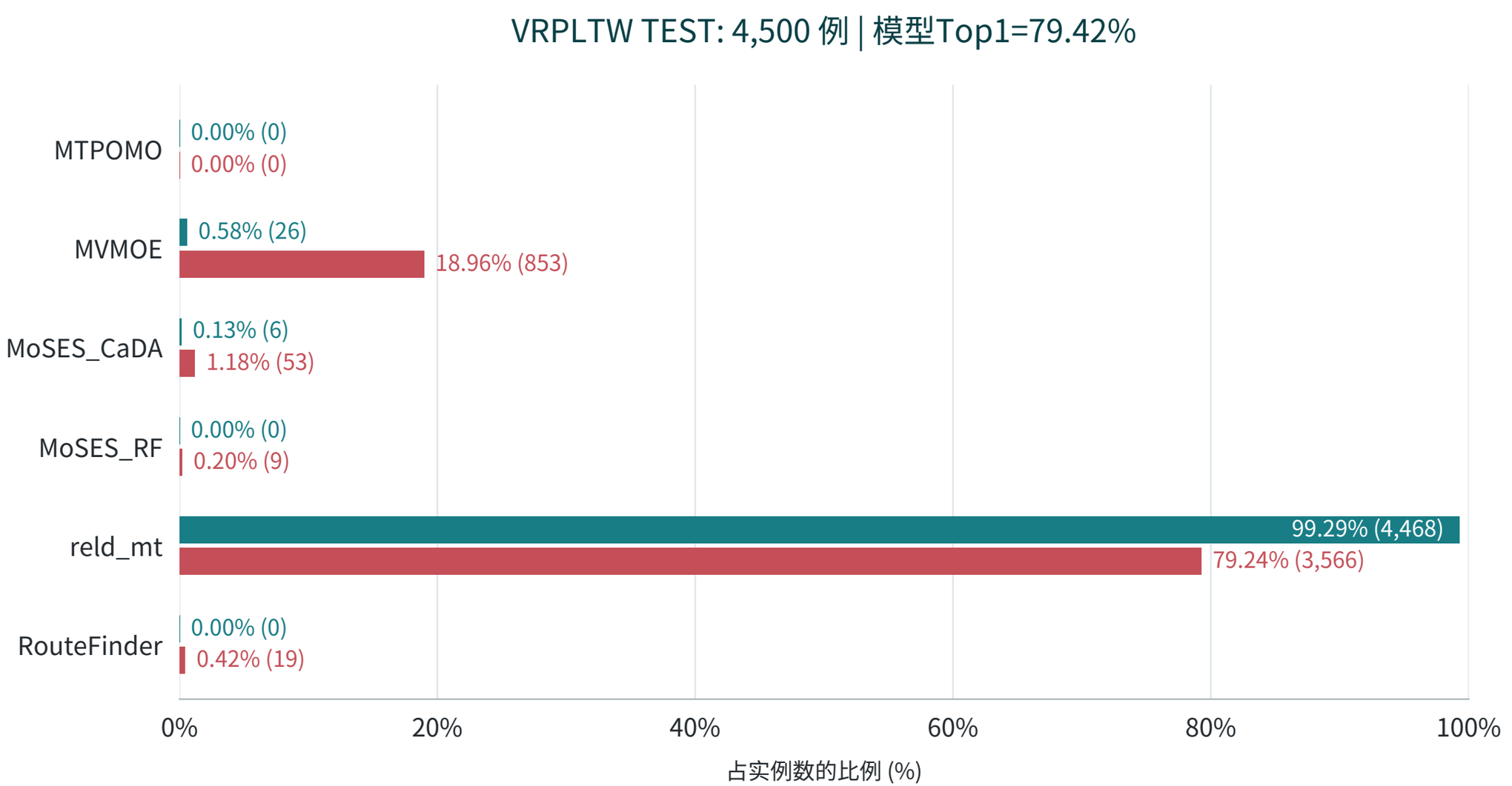
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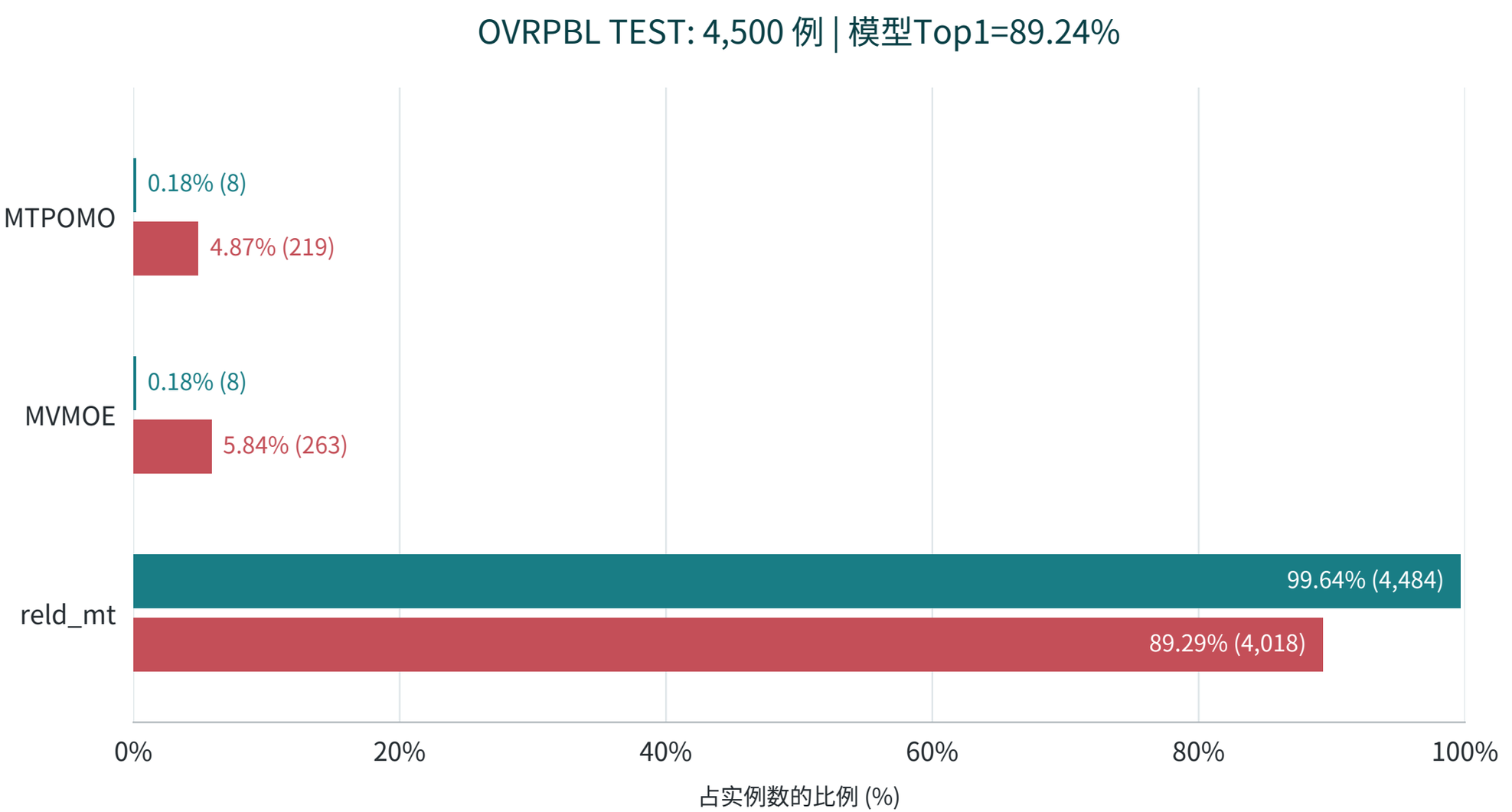
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R101/B | OVRPBL | 方法分布

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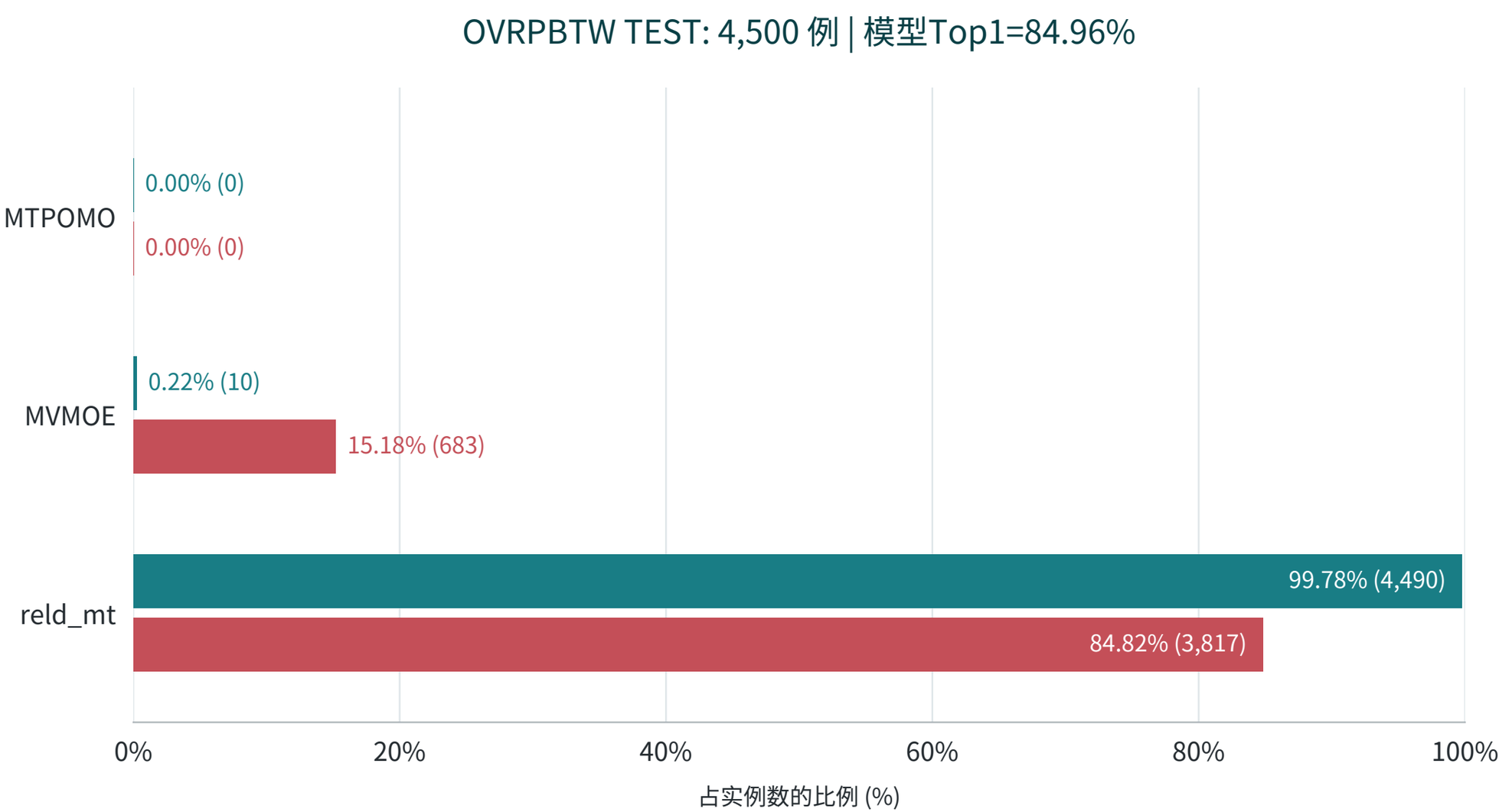
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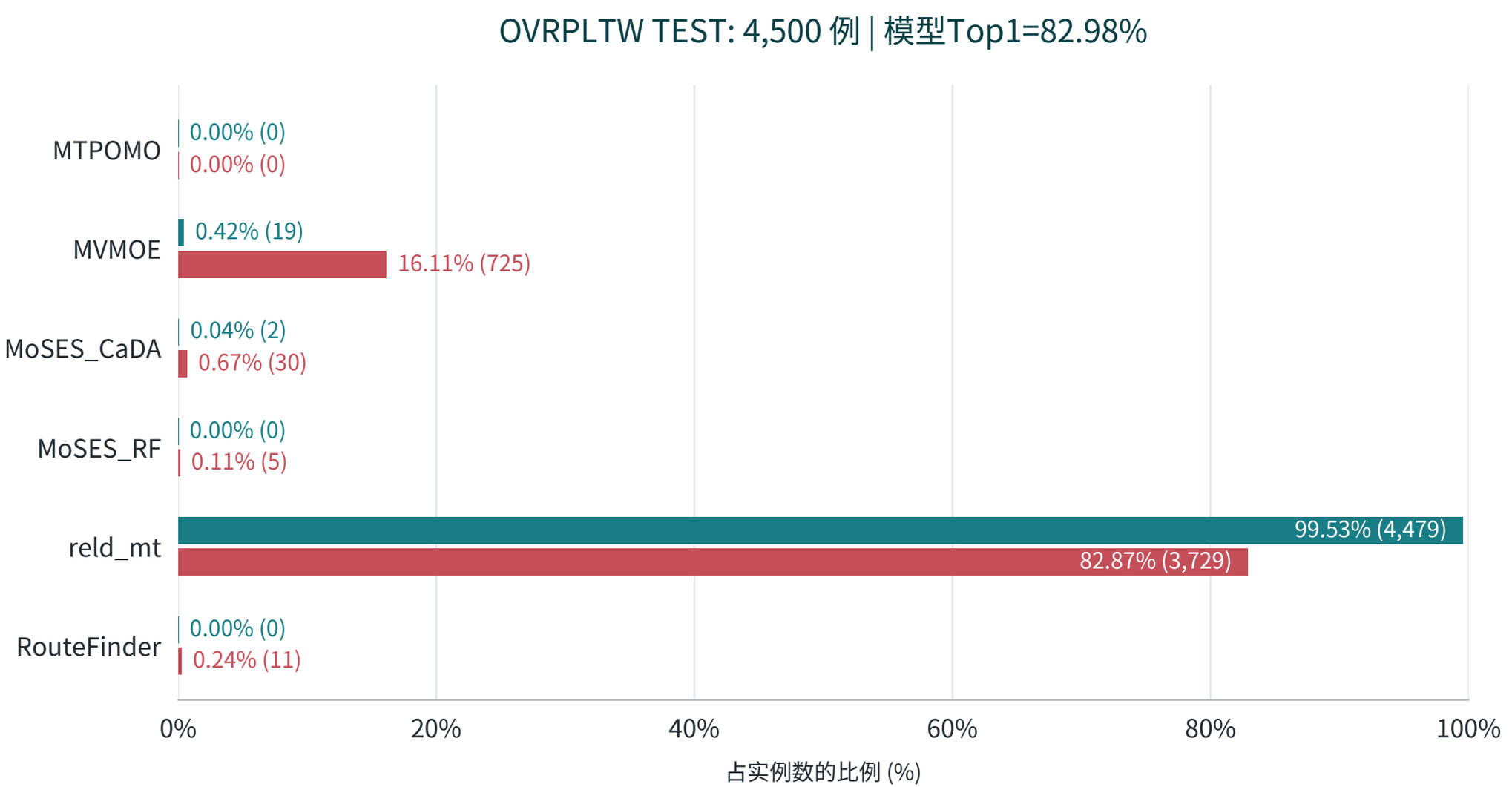
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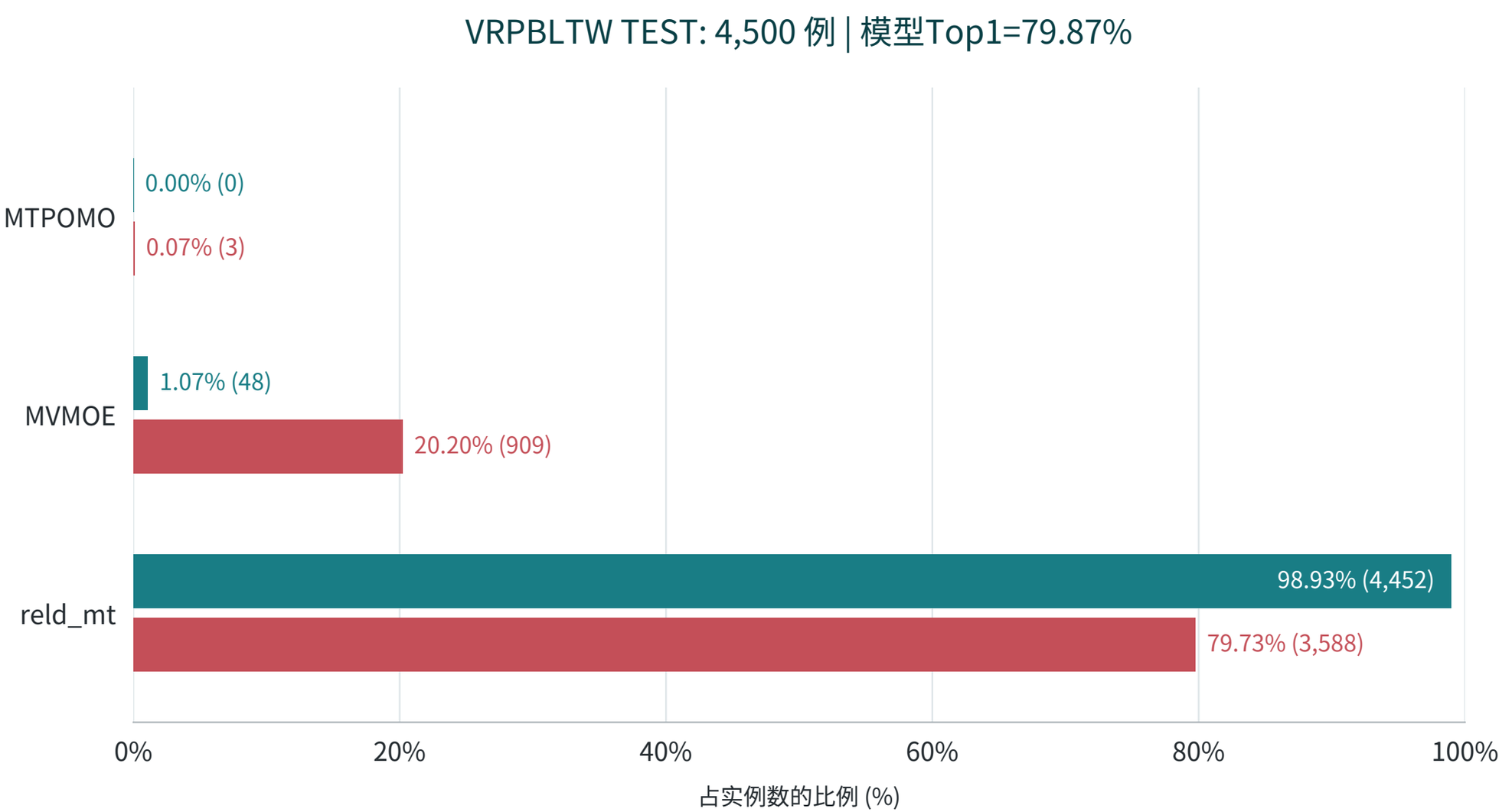
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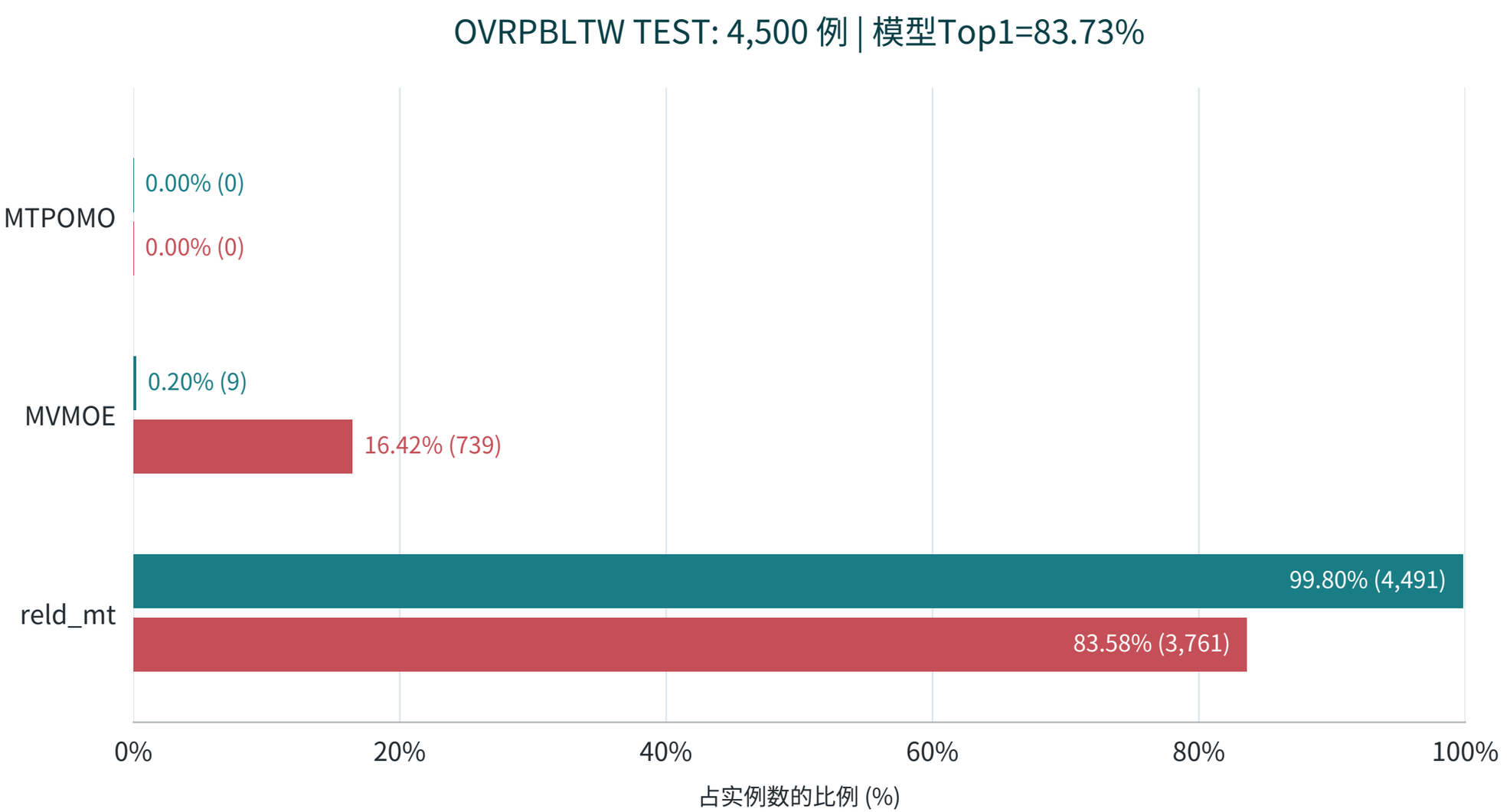
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